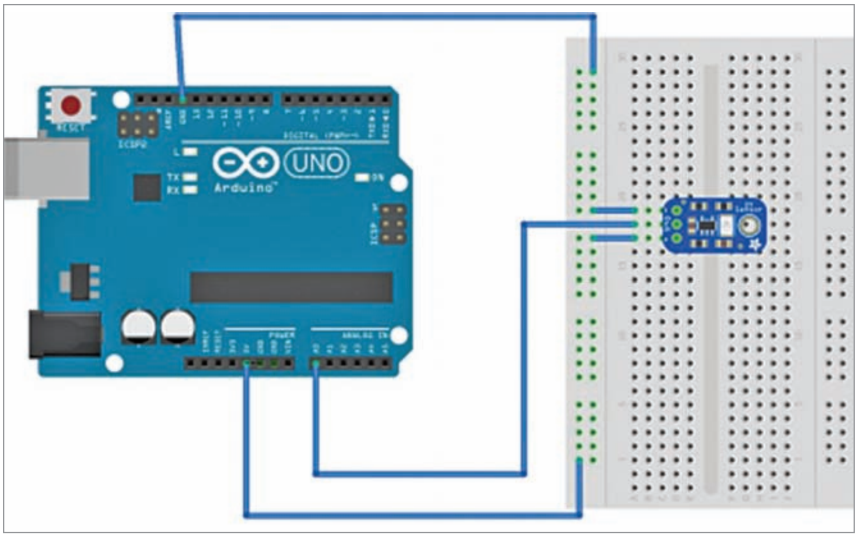

Fig. 5: Wiring of the proposed UV sensor on the breadboard

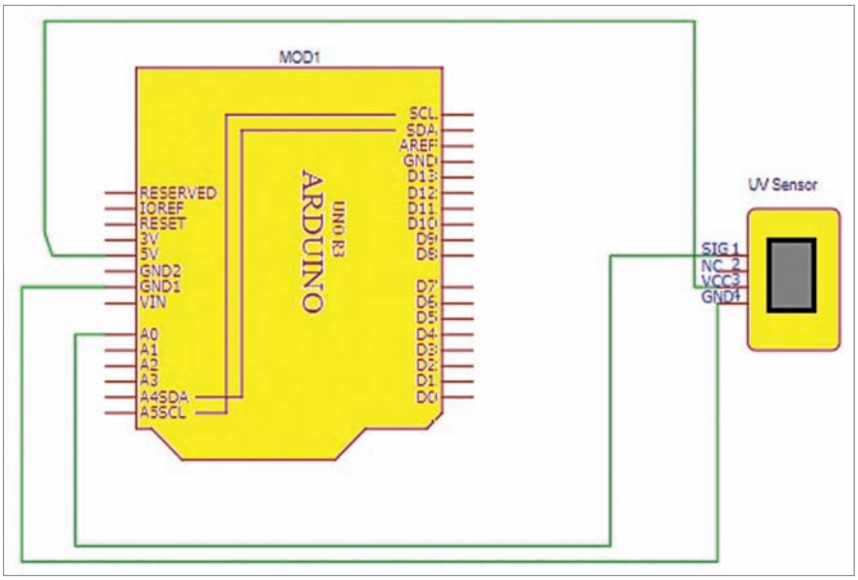

Fig. 6: Circuit diagram of the proposed system

Fig. 3 shows the UV index, the output voltage (in mV) and the corresponding analogue value. GUVA-S12SD is a gallium nitride material based Schottky-type photodiode, optimised for photovoltaic mode operation. It is followed by an op-amp IC SGM8521. It is a rail-to-rail input and output voltage feedback amplifier. Op-amps that use the complete span between negative and positive supply voltages are called rail-to-rail op-amps. Normal op-amps have a maximum voltage swing and cannot use the whole span of power supply. Rail-to-rail op-amps can use the complete span of power supply, thereby increasing the available signal range. This op-amp has a wide input common-mode voltage range and output voltage swing, and takes

EFY Note

The source code of this project is available for free download at *www.electronicsforu.com*

```
uvray_geetali_saha
void setup()
{
 Serial.begin(9600);
}
void loop()
{
 float sensorVoltage;
 float sensorValue;
 sensorValue = analogRead(A0);
 sensorVoltage = sensorValue/1024*5.0;
 Serial.print("sensor reading = ");
 Serial.print(sensorValue);
 Serial.print(" sensor voltage = ");
 Serial.print(sensorVoltage);
 Serial.println(" V");
 delay(10000);
}
```

Fig. 7: Screenshot of the source code

the minimum operating supply voltage down to 2.1V and the maximum recommended supply voltage is 5.5V. Besides, SGM8521 provides 150kHz bandwidth at a low current consumption of 4.7µA. Fig. 4 covers most of these aspects in maximum detail.

All the components needed in this project are listed in Table 1 and circuit.

Table 1 and circuit

Table 1 and circuit diagram of the project is shown in Fig. 6. After getting the components, connect them according to the wiring diagram (Fig. 5). Pin connections for both the UV sensor and Arduino are described in Table 2.

Install the latest version of Arduino IDE and connect the Arduino to USB. Now create the program for reading sensor data, where in the loop function we check the analogue value of the sensor. Fig. 7 shows the screen shot of the source code.

Testing

When the light rays emitted by any source fall on the photodiode of the UV sensor, they generate a value. This sensor value is read using the analogue read command. Corresponding voltage is evaluated, and both the values are displayed on the serial monitor. The same sensor is